

Abdelrahman Abdallah

Ph.D. in Computer Science · Information Retrieval & Natural Language Processing · University of Innsbruck, Austria

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RESEARCH SUMMARY

My research lies at the intersection of **large language models and information retrieval**, with a focus on **LLM post-training**, retrieval and reranking systems, multimodal retrieval, and **LLM-based agents**. I build new retrievers, rerankers, and multimodal models, alongside large-scale evaluation benchmarks and widely used open-source tooling (Rankify, RankArena, HintEval) for the community. As an independent researcher, I aim to advance reasoning-driven, time-aware, and multimodal retrieval — supervising students and securing competitive funding.

EDUCATION

Ph.D. in Computer Science — *University of Innsbruck, Austria* Oct 2022 – Sep 2026

- Graduated with the highest grade, *Sehr gut* (1, excellent). Advisor: Prof. Adam Jatowt. Focus: information retrieval, document reranking, retrieval-augmented generation, temporal QA.

M.S. in Data Science & Machine Learning — *Satbayev University, Almaty, Kazakhstan* Sep 2019 – Jul 2021

- Advisor: Daniyar Nurseitov. Thesis: Handwritten Recognition and Table Detection using Deep Learning. Awarded Best Master Thesis.

B.S. in Computer Science — *Assiut University, Egypt* Sep 2011 – Jul 2015

- Summa Cum Laude — ranked 2nd in the department.

ACADEMIC APPOINTMENTS

Research Assistant — *University of Innsbruck (DiSC)* Oct 2022 – Present

- Knowledge extraction and information retrieval from unstructured documents; reranking and RAG systems.
- Creator/maintainer of the Rankify framework and lead developer of the RankArena platform. Host: Prof. Adam Jatowt.

Visiting Machine Learning Researcher — *Università Ca' Foscari Venezia, Italy* Mar 2024 – Aug 2024

- Graph neural networks and random forests for climate-change tasks. Host: Sebastiano Vascon.

Machine Learning Engineer (part-time / remote) — *DISCO App* Aug 2021 – May 2025

- Built and improved an OCR system for receipt extraction and classification; app shipped on Google Play and the App Store.

Machine Learning Engineer (remote) — *KMG Engineering* Aug 2021 – Jan 2022

- Image inpainting with GANs. Host: Daniyar Nurseitov.

Undergraduate Research Assistant — *Satbayev University, Kazakhstan* Nov 2019 – Jul 2021

- National Open Research Laboratory — deep learning for Kazakh and Russian handwritten-text databases.

Research & Teaching Assistant — *Assiut University, Egypt* Jul 2016 – Jun 2019

- Teaching assistant for CS courses; research in machine learning and computer vision.

GRANTS & FUNDING

- AI Enabled Sustainability Jurisdiction Demonstrator — Austrian Research Promotion Agency (FFG), Project No. 915229.
- *[under review]* ROAR: Recall-Oriented Agent for Comprehensive IR — FWF stand-alone proposal.

HONORS & AWARDS

- Ph.D. awarded with the highest grade, *Sehr gut* (1), University of Innsbruck (2026).
- Best Master Thesis, Satbayev University (2021).
- Satbayev University Graduate Scholarship (2019).

RESEARCH SOFTWARE & OPEN SOURCE

Rankify — *Creator & Maintainer*

- Comprehensive Python toolkit for retrieval, reranking, and RAG with automated metrics and human feedback. 650+ GitHub stars; accepted at ACL 2026 System Demonstrations.

RankArena — *Lead Developer*

- Unified platform to benchmark retrieval and reranking models against state-of-the-art baselines with human and LLM feedback (CIKM 2025).

HintEval — *Co-author*

- Open-source Python toolkit for hint generation and hint evaluation. Accepted at EMNLP 2026 System Demonstrations.

PUBLICATIONS

*Author name in bold. * denotes equal contribution.*

Peer-Reviewed Conference Proceedings

- [C34] Mohamed Mahmoud, Mostafa Farouk Senussi, Mahmoud Abdalla, Mahmoud SalahEldin Kasem, **Abdelrahman Abdallah**, Hyun-Soo Kang. “MEMORA: A Memory-Enhanced Multimodal Committee Reranking Agent for Reasoning-Intensive Retrieval.” EMNLP, 2026.
- [C33] Mohammed Ali, **Abdelrahman Abdallah**, Adam Jatowt. “SustainableQA: A Comprehensive Question Answering Dataset for Corporate Sustainability and EU Taxonomy Reporting.” EMNLP, 2026.
- [C32] **Abdelrahman Abdallah**, Mohammed Ali, Muhammad Abdul-Mageed, Adam Jatowt. “TEMPO: Realistic Multi-Domain Benchmark for Temporal Reasoning-Intensive Retrieval.” EMNLP, 2026.
- [C31] **Abdelrahman Abdallah**, Mahmoud Abdalla, Mohammed Ali, Adam Jatowt. “Argus-Retriever: Vision-LLM Late-Interaction Retrieval with Region-Aware Query-Conditioned MoE for Visual Document Retrieval.” EMNLP, 2026.
- [C30] **Abdelrahman Abdallah**, Mohammed Ali, Bhawna Piryani, Mahmoud Abdalla, Adam Jatowt. “Large Language Models Systematically Favor Popular Options: Evidence and Mitigation Across MCQs.” EMNLP, 2026.
- [C29] **Abdelrahman Abdallah**, Mohammed Ali, Adam Jatowt. “SmallReason-ColBERT: An Ultra-Small Late-Interaction Retriever for Reasoning Intensive Retrieval.” EMNLP, 2026.
- [C28] **Abdelrahman Abdallah**, AbdelRahim A. Elmadany, Sameh Al Natour, Hasan Cavusoglu, Adam Jatowt, Muhammad Abdul-Mageed. “MoCA-Agent: A Market-of-Claims Code Agent for Financial and Numerical Reasoning.” EMNLP, 2026.
- [C27] Mahmoud Abdalla*, **Abdelrahman Abdallah***, Shaimaa Sedek, Adam Jatowt. “OBLIQ-IR: Training a Dense Retriever for Oblique Queries.” EMNLP, 2026.
- [C26] Jamie Holdcroft, **Abdelrahman Abdallah**, Adam Jatowt. “The Magnitude Mirage: Rethinking Confidence for Reasoning-Intensive Retrieval.” EMNLP, 2026.
- [C25] Mohammed Ali, **Abdelrahman Abdallah**, Adam Jatowt. “REGREACT: Self-Correcting Multi-Agent Pipelines for Structured Regulatory Information Extraction.” Findings of EMNLP, 2026.
- [C24] Jamshid Mozafari, Bhawna Piryani, **Abdelrahman Abdallah**, Adam Jatowt. “HintEval: An Open-Source Python Toolkit for Hint Generation and Hint Evaluation.” EMNLP System Demonstrations, 2026.
- [C23] Jamie Holdcroft, **Abdelrahman Abdallah**, Adam Jatowt. “Difficulty-Gated Fusion of Reasoning Views for Temporal Retrieval.” CIKM (Short Papers), 2026.
- [C22] Laurin Wischounig*, **Abdelrahman Abdallah***, Adam Jatowt. “Negative Sampling Techniques in Dense Retrieval: A Survey.” Findings of EACL, 2026.
- [C21] **Abdelrahman Abdallah**, Mohamed Darwish Mounis, Mahmoud Abdalla, Mahmoud SalahEldin Kasem, Mostafa Farouk Senussi, Mohamed Mahmoud, Mohammed Ali, Adam Jatowt, Hyun-Soo Kang. “MM-BRIGHT: A Multi-Task Multimodal Benchmark for Reasoning-Intensive Retrieval.” KDD, 2026.
- [C20] **Abdelrahman Abdallah**, Bhawna Piryani, Jamshid Mozafari, Andreas Herzinger, Jamie Holdcroft, Adam Jatowt. “Rankify: A Comprehensive Python Toolkit for Retrieval, Re-Ranking, and Retrieval-Augmented Generation.” ACL System Demonstrations, 2026.
- [C19] **Abdelrahman Abdallah**, Mohammed Ali, Bhawna Piryani, Adam Jatowt. “BracketRank: Large Language Model Document Ranking via Reasoning-based Competitive Elimination.” ACL, 2026.

- [C18] Bhawna Piryani, **Abdelrahman Abdallah**, Jamshid Mozafari, Avishek Anand, Adam Jatowt. "It's High Time: A Survey of Temporal Information Retrieval and Question Answering." ACL, 2026.
- [C17] Mohammed Ali, **Abdelrahman Abdallah**, Amit Agarwal, Hitesh Laxmichand Patel, Adam Jatowt. "RECOR: Reasoning-focused Multi-turn Conversational Retrieval Benchmark." Findings of ACL, 2026.
- [C16] **Abdelrahman Abdallah**, Jamie Holdcroft, Mohammed Ali, Adam Jatowt. "Are LLM-Based Retrievers Worth Their Cost? An Empirical Study of Efficiency, Robustness, and Reasoning Overhead." SIGIR, 2026.
- [C15] **Abdelrahman Abdallah**, Bhawna Piryani, Jonas Wallat, Avishek Anand, Adam Jatowt. "TempRetriever: Fusion-based Temporal Dense Passage Retrieval for Time-Sensitive Questions." WSDM, 2026.
- [C14] Bhawna Piryani, Jamshid Mozafari, **Abdelrahman Abdallah**, Antoine Doucet, Adam Jatowt. "Evaluating Robustness of LLMs in Question Answering on Multilingual Noisy OCR Data." CIKM, 2025.
- [C13] **Abdelrahman Abdallah**, Mahmoud Abdalla, Bhawna Piryani, Jamshid Mozafari, Mohammed Ali, Adam Jatowt. "RerankArena: A Unified Platform for Evaluating Retrieval, Reranking and RAG with Human and LLM Feedback." CIKM, 2025.
- [C12] Raphael Gruber, **Abdelrahman Abdallah**, Michael Färber, Adam Jatowt. "ComplexTempQA: A 100M Dataset for Complex Temporal Question Answering." EMNLP, 2025.
- [C11] **Abdelrahman Abdallah**, Bhawna Piryani, Jamshid Mozafari, Mohammed Ali, Adam Jatowt. "How Good are LLM-based Rerankers? An Empirical Analysis of State-of-the-Art Reranking Models." Findings of EMNLP, 2025.
- [C10] **Abdelrahman Abdallah**, Jamshid Mozafari, Bhawna Piryani, Adam Jatowt. "DeAR: Dual-Stage Document Reranking with Reasoning Agents via LLM Distillation." Findings of EMNLP, 2025.
- [C9] Jamshid Mozafari, **Abdelrahman Abdallah**, Bhawna Piryani, Adam Jatowt. "Wrong Answers Can Also Be Useful: PlausibleQA — A Large-Scale QA Dataset with Answer Plausibility Scores." SIGIR, 2025.
- [C8] Jonas Wallat, **Abdelrahman Abdallah**, Adam Jatowt, Avishek Anand. "A Study into Investigating Temporal Robustness of LLMs." Findings of ACL, 2025.
- [C7] **Abdelrahman Abdallah**, Jamshid Mozafari, Bhawna Piryani, Adam Jatowt. "ASRank: Zero-Shot Re-Ranking with Answer Scent for Document Retrieval." Findings of NAACL, 2025.
- [C6] **Abdelrahman Abdallah**, Jamshid Mozafari, Bhawna Piryani, Mohammed M. Abdelgwad, Adam Jatowt. "DynRank: Improve Passage Retrieval with Dynamic Zero-Shot Prompting Based on Question Classification." COLING, 2025.
- [C5] **Abdelrahman Abdallah**, Mahmoud SalahEldin Kasem, Ibrahim Abdelhalim, Norah Saleh Alghamdi, Sohail Contractor, Ayman El-Baz. "CascadePLS-ViT: Cascade with Patch-Level Self-Supervised Vision Transformers for Breast Cancer Classification in Mammography." ISBI, 2025.
- [C4] **Abdelrahman Abdallah**, Mahmoud Kasem, Mahmoud Abdalla, Mohamed Mahmoud, Mohamed Elkasaby, Yasser Elbendary, Adam Jatowt. "ArabicaQA: A Comprehensive Dataset for Arabic Question Answering." SIGIR, 2024.
- [C3] Mahmoud SalahEldin Kasem*, **Abdelrahman Abdallah***, Ibrahim Abdelhalim*, Norah Saleh Alghamdi, Sohail Contractor, Ayman El-Baz. "IHRRB-DINO: Identifying High-Risk Regions of Breast Masses in Mammogram Images Using Data-Driven Instance Noise." MICCAI, 2024.
- [C2] Bhawna Piryani, **Abdelrahman Abdallah**, Jamshid Mozafari, Adam Jatowt. "Detecting Temporal Ambiguity in Questions." Findings of EMNLP, 2024.
- [C1] Jamshid Mozafari, **Abdelrahman Abdallah**, Bhawna Piryani, Adam Jatowt. "Exploring Hint Generation Approaches for Open-Domain Question Answering." Findings of EMNLP, 2024.

Journal Articles

- [J9] Ibrahim Abdelhalim, Yassir Almalki, **Abdelrahman Abdallah**, et al. "A Deep Learning Framework for Accurate Mammographic Mass Classification Using Local Context Attention Module." Medical Physics, 2025.
- [J8] **Abdelrahman Abdallah**, Mahmoud SalahEldin Kasem, Ibrahim Abdelhalim, Norah Saleh Alghamdi, Ayman El-Baz. "Improving BI-RADS Mammographic Classification With Self-Supervised Vision Transformers and Cascade Learning." IEEE Access, 2025.
- [J7] Mahmoud Abdalla, Mahmoud SalahEldin Kasem, Mohamed Mahmoud, Bilel Yagoub, Mostafa Farouk Senussi, **Abdelrahman Abdallah**, Seung Hun Kang, Hyun Soo Kang. "ReceiptQA: A Question-Answering Dataset for Receipt Understanding." Mathematics, 2025.
- [J6] **Abdelrahman Abdallah**, Daniel Eberharter, Zoe Pfister, Adam Jatowt. "A Survey of Recent Approaches to Form Understanding in Scanned Documents." Artificial Intelligence Review, 2024.

- [J5] Mahmoud Salaheldin Kasem, **Abdelrahman Abdallah**, et al. "Deep Learning for Table Detection and Structure Recognition: A Survey." ACM Computing Surveys, 2024.
- [J4] Daniyar B. Nurseitov, Galymzhan Abdimanap, **Abdelrahman Abdallah**, et al. "ROSID: Remote Sensing Satellite Data for Oil Spill Detection on Land." Engineered Science, 2024.
- [J3] **Abdelrahman Abdallah**, Bhawna Piryani, Adam Jatowt. "Exploring the State of the Art in Legal QA Systems." Journal of Big Data, 2023.
- [J2] Nazgul Toiganbayeva, Mahmoud Kasem, Galymzhan Abdimanap, Kairat Bostanbekov, **Abdelrahman Abdallah**, Anel Alimova, Daniyar Nurseitov. "KOHTD: Kazakh Offline Handwritten Text Dataset." Signal Processing: Image Communication, 2022.
- [J1] **Abdelrahman Abdallah**, Alexander Berendeyev, Islam Nuradin, Daniyar Nurseitov. "TNCR: Table Net Detection and Classification Dataset." Neurocomputing, 2022.

Workshop Papers

- [W1] Mahmoud Abdalla, Mahmoud SalahEldin Kasem, Mohamed Mahmoud, Mostafa Farouk Senussi, **Abdelrahman Abdallah**, Hyun-Soo Kang. "HIVE: Query, Hypothesize, Verify — An LLM Framework for Multimodal Reasoning-Intensive Retrieval." CVPR 2026 Workshop GRAIL-V, 2026.
- [W2] Mohamed Darwish Mounis, Mohamed Mahmoud, Shaimaa Sedek, Mahmoud Abdalla, Mahmoud SalahEldin Kasem, **Abdelrahman Abdallah**, Hyun-Soo Kang. "BRIDGE: Multimodal-to-Text Retrieval via Reinforcement-Learned Query Alignment." CVPR 2026 Workshop GRAIL-V, 2026.
- [W3] Mahmoud SalahEldin Kasem, Mohamed Mahmoud, Mostafa Farouk Senussi, Mahmoud Abdalla, **Abdelrahman Abdallah**, Hyun-Soo Kang. "MARVEL: Multimodal Adaptive Reasoning-intensive Expand-rerank and retrieval." CVPR 2026 Workshop GRAIL-V, 2026.

INVITED TALKS

Dual-Stage Document Reranking with Reasoning Agents via LLM Distillation — *Host: Prof. Han Xiao* Nov 2025

Advancing IR: zero-shot re-ranking and dynamic prompting for passage retrieval — *ifi Lunchtime Seminar, UIBK* Apr 2025

Generate-Retrieve-Generate: A Novel Approach to Open-Domain QA — *ifi Lunchtime Seminar, UIBK* Mar 2024

TEACHING

University of Innsbruck

- *Lectures (VU)*: Advanced Data Engineering and Analytics — Understanding Natural Language — **delivered 5 lectures** (Summer 2025 & Summer 2026).
- *Practical sessions / exercises (PS)*: Machine Learning (Summer 2024); Data and Probabilities (Summer 2024, Winter 2025/26); Computer Networks and Internet Technology (Winter 2021/22, Winter 2025/26).

Assiut University — Teaching Assistant: Web Programming, Introduction to Programming (C++), Data Structures, Machine Learning, Project Management, Network Management.

STUDENT SUPERVISION & MENTORING

- Andreas Herzinger (MSc, Fall 2024 – Summer 2025) — co-authored (Rankify).
- Baraa Ahmed (MSc, Winter 2024 – present) — co-authored.

PROFESSIONAL SERVICE

Shared Task Organization

- Organizer, SemEval-2027 Task 1: RETECO — A Shared Task on Reasoning-Oriented Retrieval. Co-organizers: Mohammed Ali, Muhammad Abdul-Mageed, Kevin Duh, and Adam Jatowt.

Conference / journal reviewer:

- ACL Rolling Review (2023–2026); ACL; EMNLP (2024); ICLR; NeurIPS; SIGIR (2024, 2025, 2026); CIKM (2024, 2025, 2026); COLING (2025); ECIR (2025).

REFERENCES

- Prof. Adam Jatowt — University of Innsbruck — Adam.Jatowt@uibk.ac.at

- Prof. Islam A. T. F. Taj-Eddin — Assiut University — itajeddin@aun.edu
- Prof. Daniyar B. Nurseitov — KMG Engineering — nurseitovdb@gmail.com